

Vincent (Dam Duc Huy) Ngo

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EDUCATION

University of Alberta

B.Sc. Computer Engineering, Co-op - GPA: 3.9/4.0

Edmonton, AB

Expected 2029

ENGINEERING EXPERIENCE

Electrical and Control Engineering Intern

May 2026 – Aug. 2026

Red Deer Polytechnic - CIM-TAC / Energy Innovation Centre

Red Deer, AB

- Architected dual-core **FreeRTOS** firmware on ESP32-S3 and bare-metal C on PIC18F24Q10 for a non-invasive sensing platform; reduced average current draw **70.4% (85.71 mA to 25.33 mA)** using GPIO ISRs, task notifications, peripheral power gating, and watchdog-based sleep scheduling.
- Replaced brittle static sensing thresholds with a rate-of-change state machine backed by a **300 s SRAM ring buffer** and EMA filtering; diagnosed queue-blocking E-paper shutdown soft-locks and implemented a queue-bypass teardown path to prevent FreeRTOS kernel panics.
- Engineered an in-circuit **Spy-Bi-Wire ROM BSL flasher** in C++/FreeRTOS on ESP32-C6 for MSP430FR6043, streaming CRC-16 validated 240-byte payloads to 20-bit FRAM at 115,200 baud; eliminated parasitic back-powering with High-Z GPIO sequencing.
- Migrated BLE firmware from Bluedroid to NimBLE, reducing static binary size by approximately **600 KB (over 35%)** and enabling an 8 MB dual-image OTA layout with automated firmware staging.
- Designed a **65 mm x 18 mm, 4-layer mixed-signal PCB** for an instrumented wheelchair curling stick, integrating ESP32-S3, 6-DoF IMU and force sensing, ideal-diode power paths, low-noise analog regulation, and 50 Hz batched BLE telemetry.
- Commissioned a **4.5 kW, 230 VAC 3-phase servo system** with dual-channel Safe Torque Off, EMI filtering, regenerative braking, and a custom isolated mixed-signal interface PCB; implemented a 100 Hz FreeRTOS PID loop regulating tether tension up to 300 lbf (1,334 N).

Flight Software & Embedded Systems Team Member

Sept. 2024 – Apr. 2025

AlbertaSat - Ex-Altas 3 CubeSat

Edmonton, AB

- Contributed 7+ months to the Ex-Altas 3 3U CubeSat OBC platform spanning **embedded Linux (Yocto/PetaLinux), RTOS, boot configuration, Rust flight software, and FPGA digital logic**, including fault-tolerant boot and watchdog concepts for LEO operations.
- Developed asynchronous Rust flight-software handlers and ground-station CLI tools, and contributed VHDL/Verilog FPGA logic for telemetry and high-speed data handling.
- Performed hardware-in-the-loop integration and ISO 8 clean-room qualification, diagnosing UART, SPI, I2C, and CAN communication faults with oscilloscopes and logic analyzers and authoring version-controlled verification documentation.

PROJECTS

Bare-Metal Real-Time Audio Spectrum Visualizer | STM32, C, CMSIS-DSP, DMA, PWM

2025

- Built a register-level STM32F103C8 pipeline using ADC + DMA, timer interrupts, and a **1024-point CMSIS-DSP FFT**; reduced FFT processing from 28 ms to **15 ms** and achieved **17 ms end-to-end latency** within a 20 KB SRAM budget.
- Generated nanosecond-accurate 800 kHz WS2812B waveforms using timer PWM while maintaining non-blocking audio acquisition; validated DMA throughput and pulse timing with GDB, a logic analyzer, and oscilloscope.

PROJECTS

- NEST Wearable Health Monitor** | *ESP32, ESP-IDF, FreeRTOS, KiCad* 2025
- Architected a dual-core ESP-IDF/FreeRTOS system separating high-rate PPG/IMU acquisition from GUI and Wi-Fi tasks, using queues, semaphores, mutexes, and SPI DMA to prevent display-induced task starvation.
 - Designed and assembled a compact 2-layer mixed-signal PCB integrating MAX30102, MPU6500, ENS160, and AHT21 sensors; validated 3.3 V rails and I2C/SPI integrity and implemented power-aware sensor duty cycling in a 3D-printed wearable prototype.
- High-Throughput Serial & BLE Telemetry Engine** | *C#, .NET 8, System.IO.Pipelines* 2026
- Architected a multi-threaded C#/.NET 8 telemetry engine using **System.IO.Pipelines** and thread-safe channels for resilient serial/BLE framing, mixed-endian decoding, checksum validation, and sustained logging without UI-thread stalls.
- Automated Embedded Build & Deployment Pipeline** | *Bash, PowerShell, WSL2, XC8* 2026
- Engineered a headless PowerShell/Bash build-and-flash pipeline around **arduino-cli** and native Linux **xc8-cc**, standardizing multi-target builds and automated OTA/in-circuit firmware staging while cutting deployment overhead by **over 60%**.
- Biomechanical Telemetry Dashboard** | *Flutter, Dart, BLE, Reactive UI* 2026
- Built a cross-platform Flutter dashboard decoding 180-byte BLE telemetry packets and plotting multi-axis kinetic data at **60 FPS**; isolated rendering with **RepaintBoundary** and managed stream lifecycles to eliminate UI frame drops during high-rate logging.

TECHNICAL SKILLS

Languages: C, C++, Embedded C, Rust, Python, C#, Bash, PowerShell, VHDL, Verilog
Embedded & Real-Time: FreeRTOS, ESP-IDF, bare-metal ARM/CMSIS, GPIO, ADC, DMA, timers, ISRs, task/queue synchronization, peripheral drivers
MCUs & Compute: STM32 (ARM Cortex-M), ESP32-S3/C6, MSP430FR6043, PIC18F24Q10, FPGA/SoC platforms
Embedded Linux & Systems: Yocto Project, PetaLinux, Buildroot, Linux/POSIX, WSL2, cross-compilation, boot/firmware utilities, multithreaded CLI tooling
Interfaces & Networking: UART/USART, SPI, I2C, CAN, BLE, Wi-Fi, RS-485/Modbus RTU, 4–20 mA, JTAG/SWD, Spy-Bi-Wire
Digital & Signal Processing: VHDL, Verilog, Vivado, CMSIS-DSP, FFT, real-time signal processing, MATLAB
PCB & Hardware: KiCad, Altium Designer, mixed-signal design, DFM/DFA, hardware bring-up, SMT assembly, soldering/rework
Debug & Lab: Git, GDB, oscilloscopes, logic analyzers, DMMs, protocol analysis, Nordic PPK2, hardware/firmware integration debugging

AWARDS & SCHOLARSHIPS

- IEEE Northern Canada Section Scholarship** 2026
- The Albert Julian Hrapko Memorial Scholarship in Computer Engineering** 2025
- The Robert Stollery Scholarship in Engineering** 2024–2025
- Third Place - Natignite 2025 Hackathon** 2025